

Class 9: Halloween Candy Mini-Project

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Today we will take a wee step back to some data we can taste and explore the correlation structure and principal components of some Halloween candy.

1. Importing data

```
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294

One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity == "1")
```

```
[1] 38
```

2. What is your favorite candy

Q3. What is your favorite candy in the dataset and what is its winpercent value?

```
candy["Swedish Fish", ]$winpercent
```

```
[1] 54.86111
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Exploratory Analysis

We can use the **skimr** package to get a quick overview of a given dataset. This can be useful for the first time you encounter a new dataset.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete	ratio	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99		
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98		
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18		

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

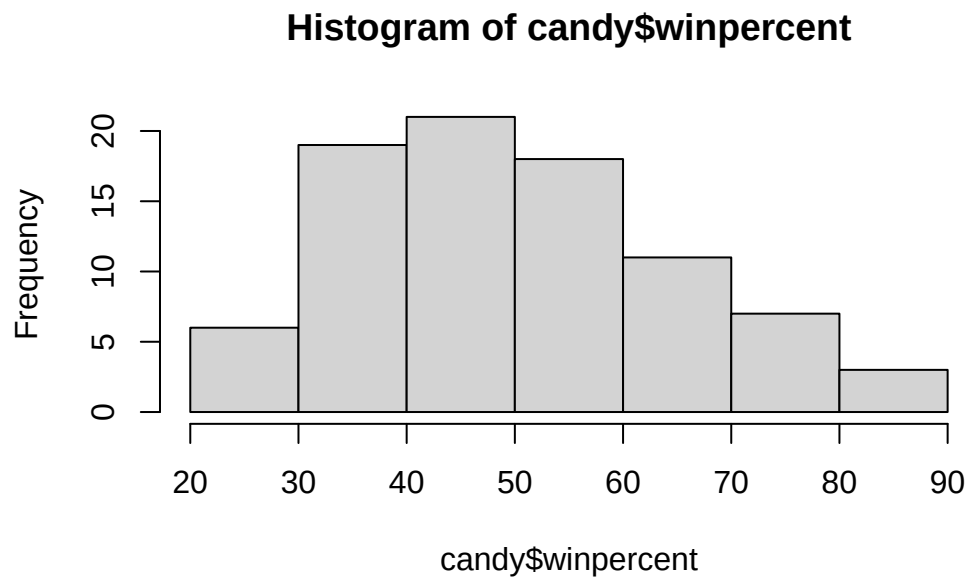
It looks like the last column `candy$winpercent` is on a different scale to all others. Looking at the mean, sd, and p values, it seems to be on a scale of 0-100 instead of 0-1.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

It's a true or false statement; if a candy has chocolate, it's value would have 1, otherwise it would have 0.

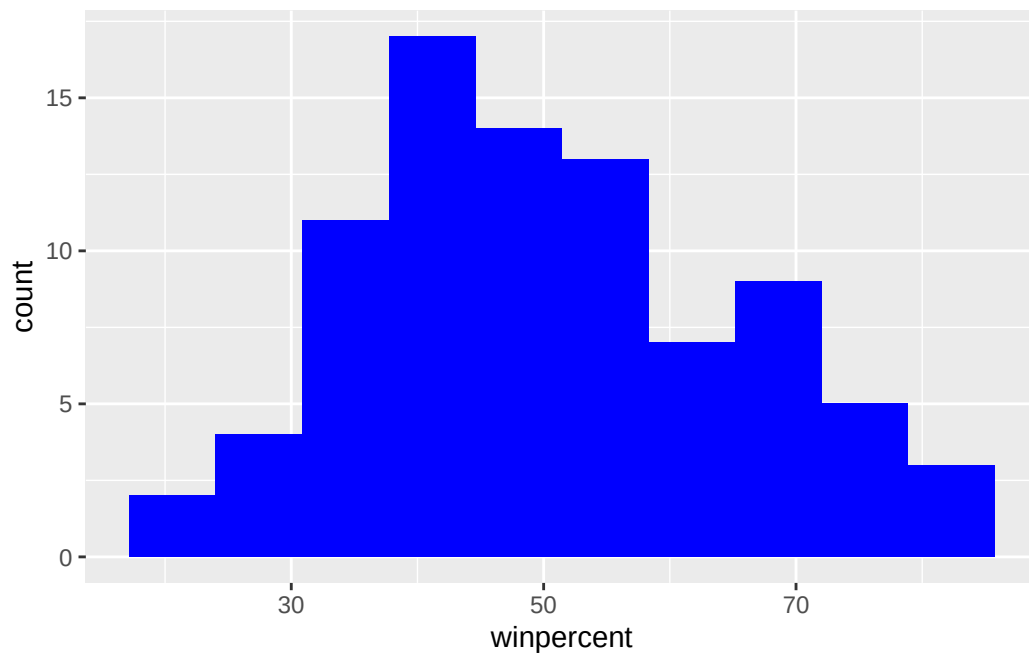
Q8. Plot a histogram of winpercent values

```
hist(candy$winpercent)
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=10, fill="blue")
```



Q9. Is the distribution of winpercent values symmetrical?

No, skewed left.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Median is below 50%, mean is on 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.inds <- candy$chocolate == 1
choc.candy <- candy[choc.inds,]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.win <- candy[as.logical(candy$fruity),]$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

higher

Q12. Is this difference statistically significant?

```
ans <- t.test(choc.win, fruit.win)
```

Yes, with a p value of 2.8713778×10^{-8} the difference is statistically significant.

3. Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

There are two related function that can help here, one is the classic `sort()` and `order()` .

```
x <- c(5, 10, 1, 4)
sort(x)
```

```
[1] 1 4 5 10
```

```
inds <- order(candy$winpercent)
head(candy[inds,], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip		0	0	0		1		0.197		0.976
Boston Baked Beans		0	0	0		1		0.313		0.511
Chiclets		0	0	0		1		0.046		0.325
Super Bubble		0	0	0		0		0.162		0.116
Jawbusters		0	1	0		1		0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
tail(candy[inds,], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

	crispedricewafer	hard bar	pluribus	sugarpercent
Snickers	0	0	1	0.546
Kit Kat	1	0	1	0.313
Twix	1	0	1	0.546
Reese's Miniatures	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0.720

	pricepercent	winpercent
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

or

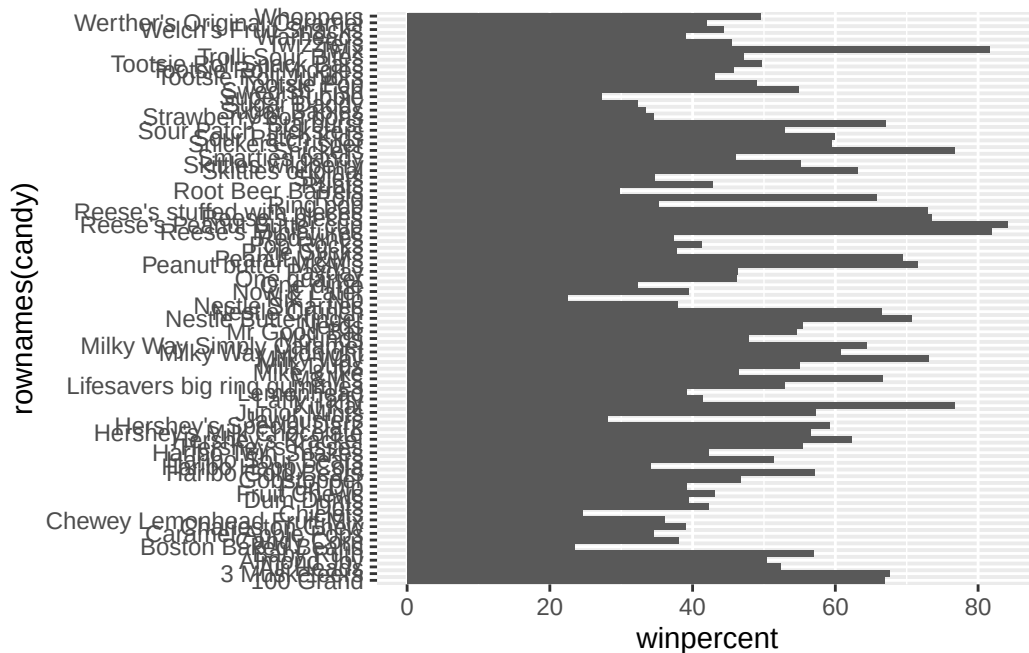
```
inds <- order(candy$winpercent, decreasing = T)
head(candy[inds,], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546
	price	percent	win	percent				
Reese's Peanut Butter cup	0.651	84.18	0.29					
Reese's Miniatures	0.279	81.86	0.26					
Twix	0.906	81.64	0.29					
Kit Kat	0.511	76.76	0.86					
Snickers	0.651	76.67	0.37					

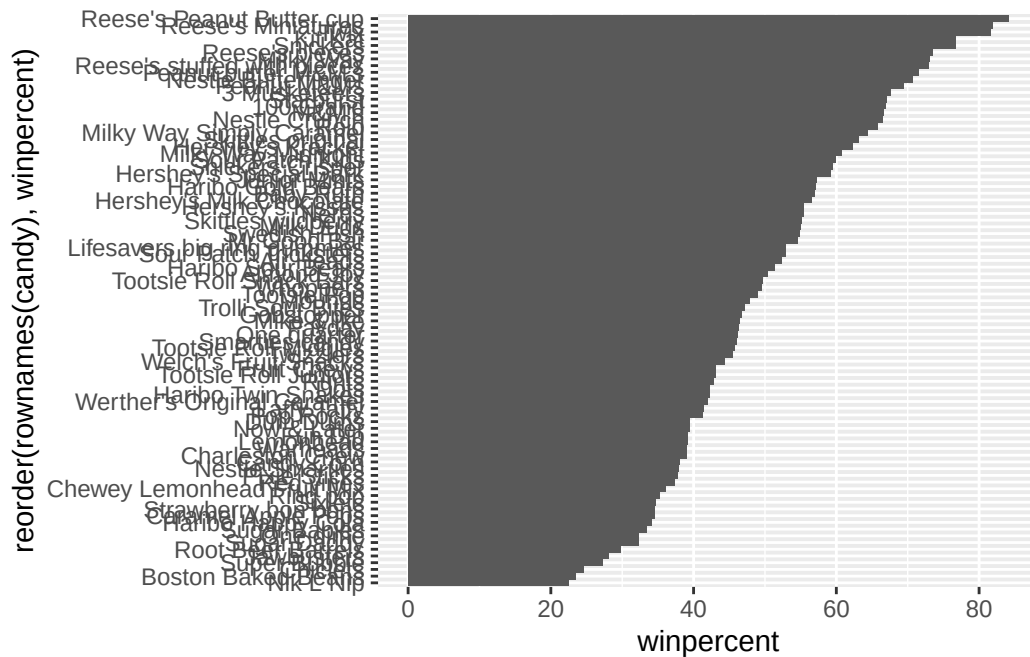
Q15. Make a first barplot of candy ranking based on winpercent values

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

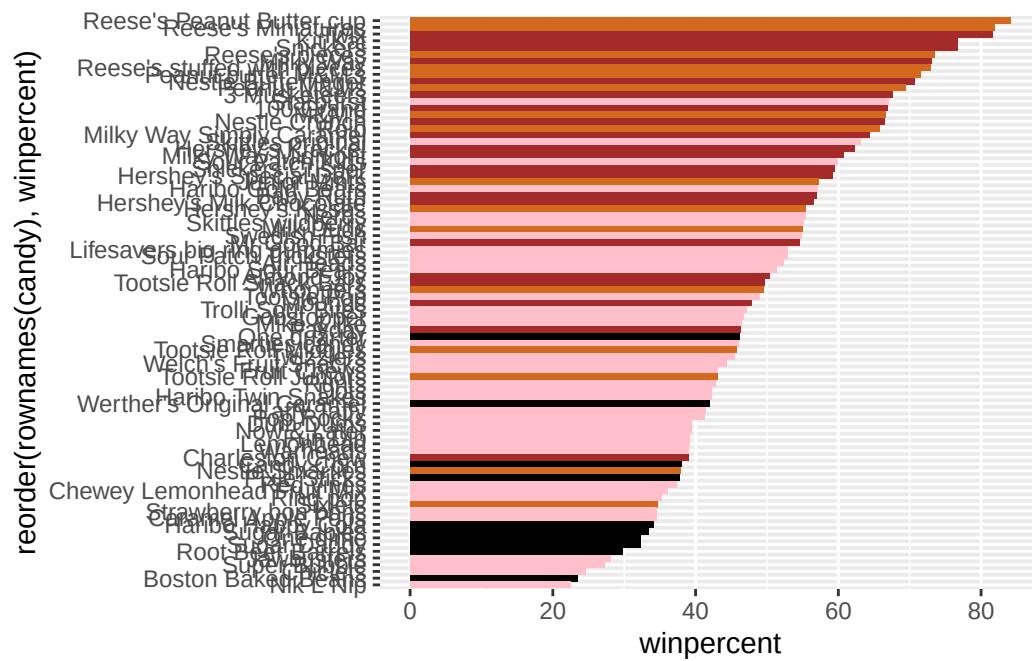

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col()
```



Here we want a custom color vector to color each bar the way we want = with chocolate and fruity candy together with whether it is a bar or not.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$fruity)] = "pink"
my_cols[as.logical(candy$bar)] = "brown"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



```
ggsave("mybarplot.png", width = 4, height = 6)
```

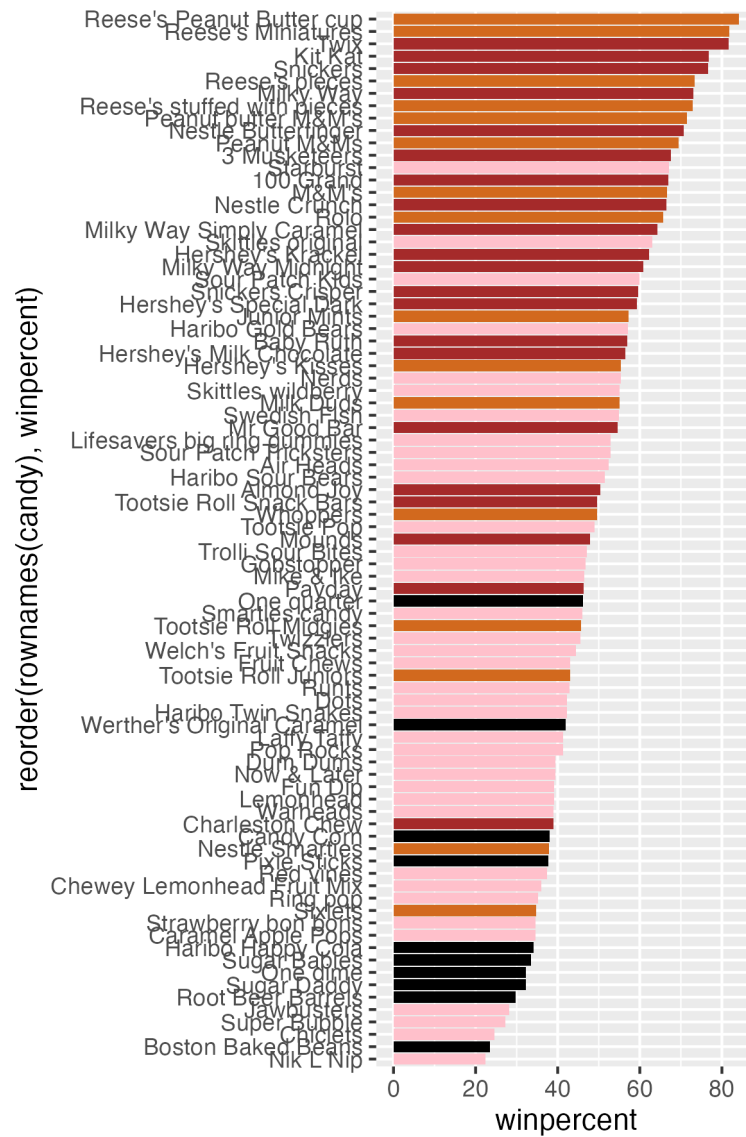


Figure 1: My silly barplot image

Q17. What is the worst ranked chocolate candy?

Sixlets

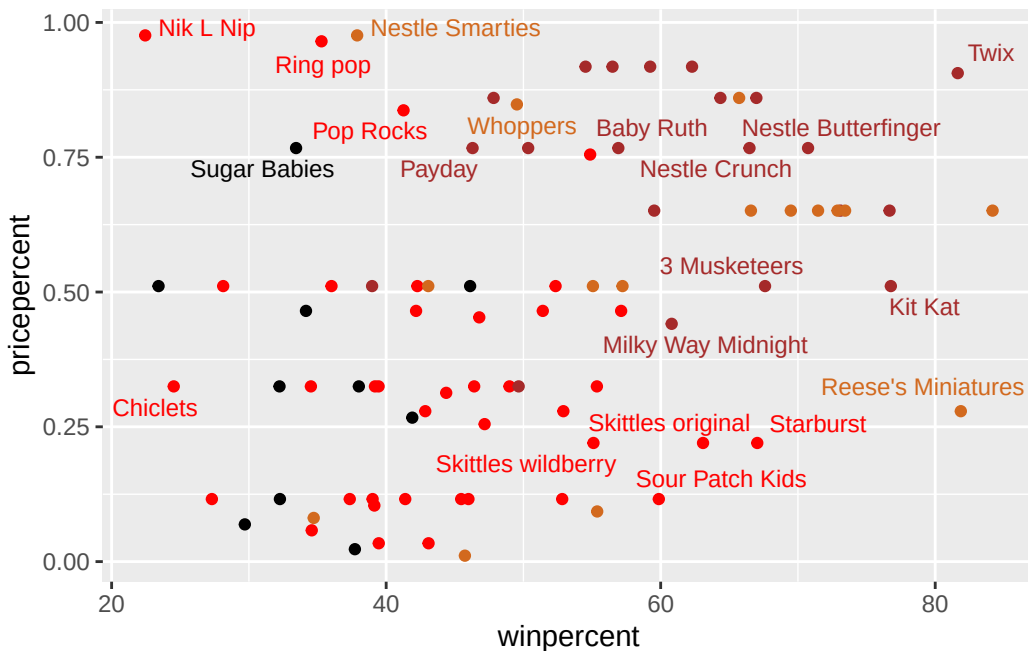
Q18. What is the best ranked fruity candy?

Starburst

4. Taking a look at pricepercent

```
library(ggrepel)
#pink is too light, lets change the colors
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "red"
# How about a plot of price vs win
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reeses Minatures seem to be the highest ranked in terms of win percent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Nik L Nip is the least popular of the top 5 most expensive candy types.

5. Correlation Structure

```
library(corrplot)
```

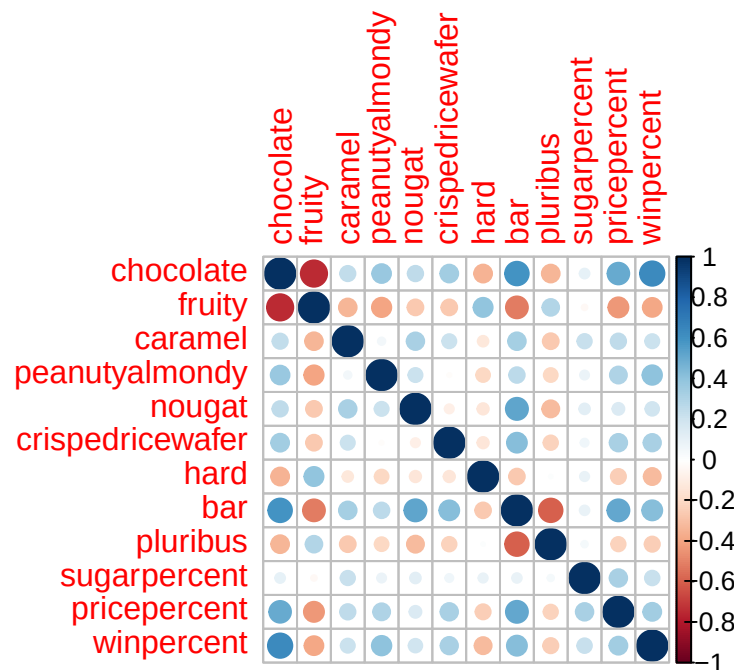
corrplot 0.95 loaded

```
cij <- cor(candy)
cij
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
chocolate	1.0000000	-0.74172106	0.24987535	0.37782357	0.25489183
fruity	-0.7417211	1.00000000	-0.33548538	-0.39928014	-0.26936712
caramel	0.2498753	-0.33548538	1.00000000	0.05935614	0.32849280
peanutyalmondy	0.3778236	-0.39928014	0.05935614	1.00000000	0.21311310
nougat	0.2548918	-0.26936712	0.32849280	0.21311310	1.00000000
crispedricewafer	0.3412098	-0.26936712	0.21311310	-0.01764631	-0.08974359
hard	-0.3441769	0.39067750	-0.12235513	-0.20555661	-0.13867505
bar	0.5974211	-0.51506558	0.33396002	0.26041960	0.52297636
pluribus	-0.3396752	0.29972522	-0.26958501	-0.20610932	-0.31033884
sugarpercent	0.1041691	-0.03439296	0.22193335	0.08788927	0.12308135
pricepercent	0.5046754	-0.43096853	0.25432709	0.30915323	0.15319643
winpercent	0.6365167	-0.38093814	0.21341630	0.40619220	0.19937530

chocolate	0.34120978	-0.34417691	0.59742114	-0.33967519
fruity	-0.26936712	0.39067750	-0.51506558	0.29972522
caramel	0.21311310	-0.12235513	0.33396002	-0.26958501
peanutyalmondy	-0.01764631	-0.20555661	0.26041960	-0.20610932
nougat	-0.08974359	-0.13867505	0.52297636	-0.31033884
crispedricewafer	1.00000000	-0.13867505	0.42375093	-0.22469338
hard	-0.13867505	1.00000000	-0.26516504	0.01453172
bar	0.42375093	-0.26516504	1.00000000	-0.59340892
pluribus	-0.22469338	0.01453172	-0.59340892	1.00000000
sugarpercent	0.06994969	0.09180975	0.09998516	0.04552282
pricepercent	0.32826539	-0.24436534	0.51840654	-0.22079363
winpercent	0.32467965	-0.31038158	0.42992933	-0.24744787
	sugarpercent	pricepercent	winpercent	
chocolate	0.10416906	0.5046754	0.6365167	
fruity	-0.03439296	-0.4309685	-0.3809381	
caramel	0.22193335	0.2543271	0.2134163	
peanutyalmondy	0.08788927	0.3091532	0.4061922	
nougat	0.12308135	0.1531964	0.1993753	
crispedricewafer	0.06994969	0.3282654	0.3246797	
hard	0.09180975	-0.2443653	-0.3103816	
bar	0.09998516	0.5184065	0.4299293	
pluribus	0.04552282	-0.2207936	-0.2474479	
sugarpercent	1.00000000	0.3297064	0.2291507	
pricepercent	0.32970639	1.0000000	0.3453254	
winpercent	0.22915066	0.3453254	1.0000000	

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

```
round(cij["chocolate", "fruity"], 2)
```

```
[1] -0.74
```

```
round(cij["pluribus", "bar"], 2)
```

```
[1] -0.59
```

Chocolate and fruity are anti-correlated, along with pluribus and bar.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and bar, along with chocolate and winpercent seem to be positively correlated.

```
round(cij["chocolate", "bar"], 2)
```

```
[1] 0.6
```

```
round(cij["chocolate", "winpercent"], 2)
```

```
[1] 0.64
```

6. Principal Component Analysis

We need to be sure to scale our input `candy` data before PCA as we have the `winpercent` column on a different scale to all others in the dataset.

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

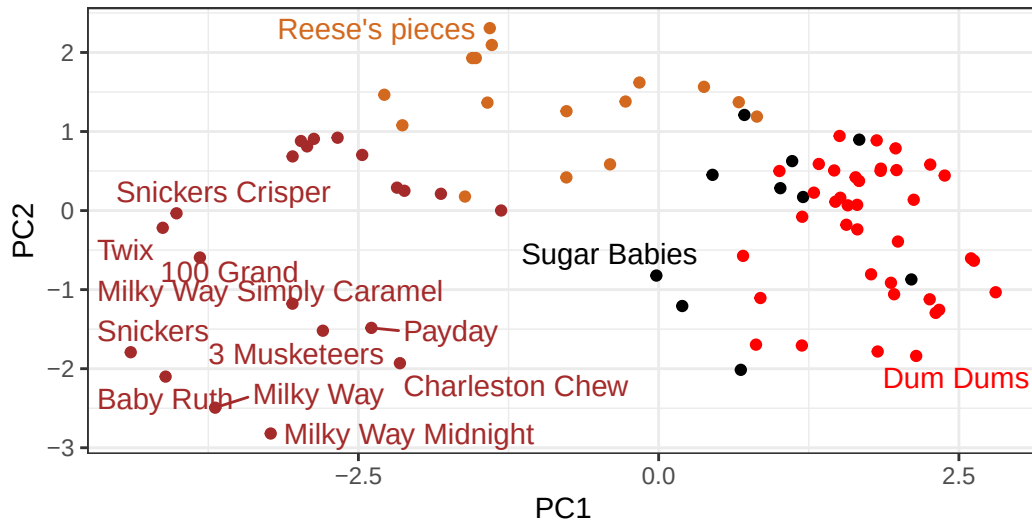
First main result figure is my “PCA plot”

```
#pca$x
ggplot(pca$x) +
  aes(PC1, PC2, label = rownames(pca$x)) +
  geom_point(col = my_cols) +
  geom_text_repel(max.overlaps = 6, col = my_cols) +
  theme_bw() +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
        caption="Data from 538")
```

Warning: ggrepel: 71 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Halloween Candy PCA Space

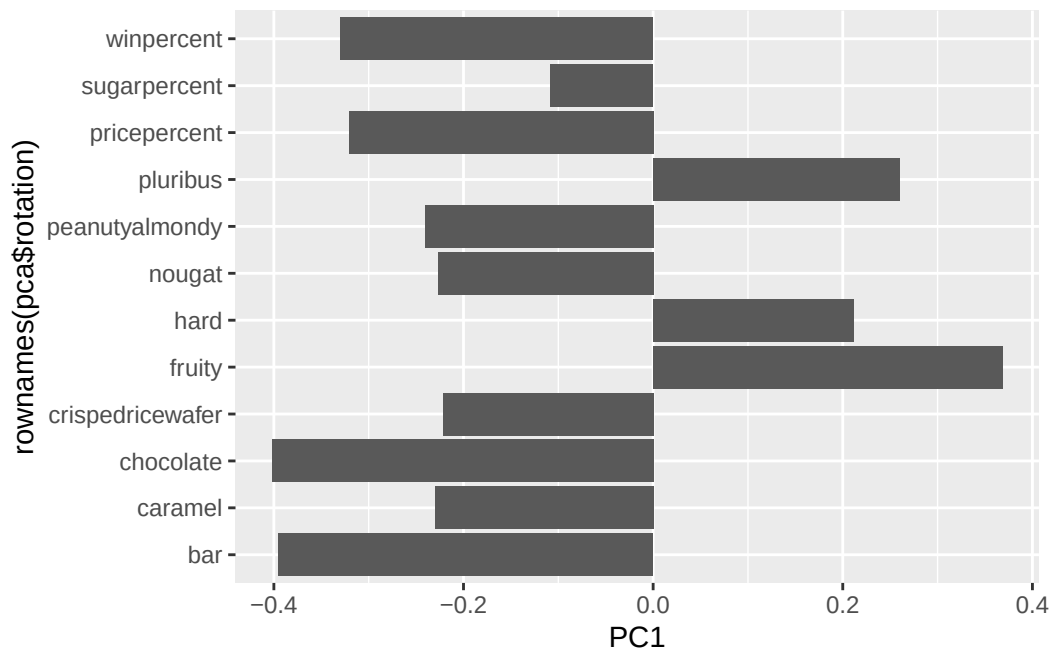
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



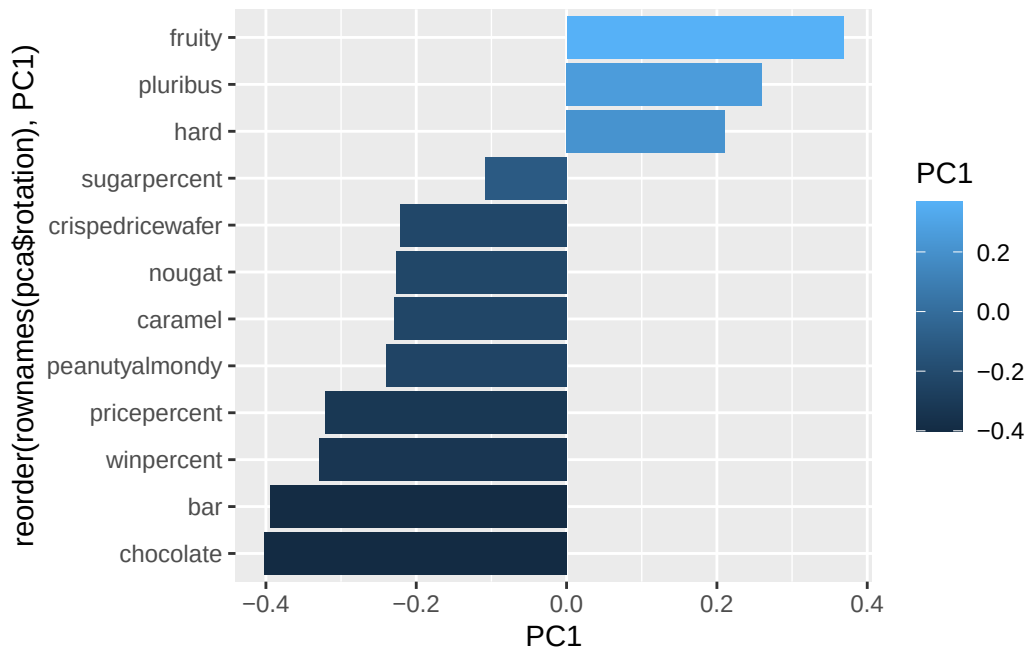
Data from 538

The second main PCA result is in the `pca$rotation`. We can plot this to generate a so-called “loadings” plot.

```
#pca$rotation  
  
ggplot(pca$rotation) +  
  aes(PC1, rownames(pca$rotation)) +  
  geom_col()
```



```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1), fill = PC1) +
  geom_col()
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

The variables that are positive in PC1 are fruity, hard, and pluribus. This makes sense as it picks up on the correlations established earlier (such as how fruity candy tend to be hard and pluribus).